

Toan Vo

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PROFESSIONAL SUMMARY

Software Engineer pursuing an M.S. in Computer Science, with 5 years of experience in software development and 3 years of hands-on experience in machine learning. Proficient in two or more general-purpose programming languages, with strong expertise in Python, C/C++, and JavaScript, and a solid foundation in data structures and algorithms. In-depth knowledge of modern software frameworks such as NodeJS and ReactJS, with experience building scalable, distributed backend and mobile systems. Experienced in developing machine learning models, with a focus on Deep Learning and Data Analysis. Skilled in popular ML frameworks such as scikit-learn, PyTorch, TensorFlow, and JAX, with a growing interest in foundational AI, ML interpretability and its applications in real-world problems.

EDUCATION

University of Maryland - College Park

M.S., Computer Science

College Park, MD

August 2026 – Present

- **Relevant courses:** Graduate NLP, Parallel Computing

University of Wisconsin - Madison

B.S., Computer Science. Certificate: Statistics, Mathematics. GPA: 3.7/4

Madison, WI

August 2022 – May 2026

- **Relevant courses:** Algorithms, Data Structures & Algorithms, Statistical Modeling, Object-Oriented Programming, Operating Systems, Database Systems, Probability, Linear Algebra, Optimization, Deep Learning

EXPERIENCE

Software Engineer Intern

A Story Technologies, Inc.

June 2026 – August 2026

Remote (Vietnam)

- Designed and developed core features of a cross-platform **React Native/Expo** mobile application for AI-guided storytelling, piloted with **30+** users at retirement communities.
- Built a **Stripe** subscription billing system with hosted checkout, webhook-driven plan state, and self-serve cancel/resume, backed by **Supabase**.
- Engineered a multi-table **Postgres** schema with ownership-chained **Row-Level Security** and folder-scoped **Storage** bucket policies.
- Built a **Flask API** integrating **OpenAI API** and **Supabase** for AI chat/voice, with **JWT** auth (**JWT**), usage logging, and signed-URL file storage.
- Engineered a distributed, real-time **WebRTC** voice AI pipeline with multimodal chat and token-level cost accounting.
- Architected scalable, stateless backend infrastructure on **AWS ECS/Fargate** with **ECR**-managed containers and centralized secrets management.

Software Engineer Intern

Viettel Solutions

June 2025 – August 2025

Ho Chi Minh City, Vietnam

- Evaluated and benchmarked multiple AI models for emerging AI-powered code-refactoring tools to automate performance optimizations across the codebase in a GPU-constrained environment.
- Designed and containerized AI model evaluation environments using **Docker**, standardizing workflows and ensuring reproducibility across platforms.
- Implemented local **LLM** inference pipelines via **Ollama**, enabling rapid model testing and benchmarking without dedicated GPU infrastructure.
- Optimized Viettel's AI Tessel **computer vision** pipeline for multi-angle grocery shelf image analysis, increasing matching accuracy by 30% and reducing end-to-end processing time by 25%.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, Java, C/C++, C#, SQL, JavaScript, TypeScript, Bash.
- **Tools & Platforms:** Git, GitHub, Unix/Linux, Supabase, Docker, AWS, Stripe, REST APIs, EAS Build.
- **Libraries & Frameworks:** React Native, Expo, React Navigation, Flask, PyTorch, TensorFlow, JAX.

RESEARCH EXPERIENCE

Undergraduate Researcher (AI semantics)

September 2025 – May 2026

Knowledge and Concepts Lab, University of Wisconsin-Madison

Madison, WI

- Conducted research on interpretability and control in neural networks, analyzing how internal representations vary across task demands.
- Evaluated semantic representations using SVM probing, dimensionality reduction, pairwise distance analysis, and multidimensional scaling (MDS).
- Ran multi-seed experiments and hyperparameter sweeps with wandb, tracking training loss and representation patterns across tasks to evaluate the stability of learned semantic structure.

Undergraduate Researcher (Machine Learning and Robotics)

May 2024 – August 2024

KuntzLab, University of Utah

Salt Lake City, UT

- Developed a novel LSTM-based model to predict tendon robot shapes from current configurations, surpassing prior state-of-the-art performance by 21%.
- Engineered a Bayesian optimization framework to automate surgical task execution, improving tissue retraction efficiency by 27% and attachment point detection accuracy by 15%.
- Contributed to lab publications and presentations, demonstrating applicability of ML techniques in robotic surgical automation.

PUBLICATIONS

PrefKit: Analyzing Preference Elicitation Methods in Qwen3 Family Models

Kaustubh Gupta, **Toan Vo**, Shinena Xiang, Siddeshwar U S, Kumar Vasagam

Apart Research Sprint, 2026.

Bayesian Retraction Optimization for Tissue Attachment Mapping in Surgical Dissection

Shing-Hei Ho, Bao Thach, **Toan Vo**, James M Ferguson, Alan Kuntz

arXiv preprint, arXiv:2607.19174, 2026.

PROJECTS

Deep Learning for Ocular Disease Classification | *Python, PyTorch*

March 2025 – April 2025

- Built deep learning models to classify 8 common ocular diseases from ODIR-5K retinal images (6,392 total), achieving 50% classification accuracy.
- Conducted systematic evaluation of ResNet, EfficientNet, and custom CNN architectures, selecting optimal networks based on validation performance.
- Integrated Grad-CAM for visual interpretability, enabling clinical validation of model predictions by highlighting optic disc and macula regions.

MiniSpark | *C*

April 2025

- Built MiniSpark, a single-node implementation of Apache Spark in C, supporting RDD DAGs (Resilient Distributed Datasets) with transformations (map, filter, join, partitionBy) and actions (count, print).
- Implemented a thread pool scheduler with locks and condition variables to materialize RDD partitions in parallel, added metrics logging for profiling, and ensured safe memory management.

Workspace Manager | *MongoDB, Javascript*

November 2023

- Developed an application to streamline workspace setup and enhance workflow efficiency for multiple users.
- Implemented MongoDB-based backend to manage settings and preferences, reducing setup time by 50%.
- Incorporated secure authentication and encrypted accounts, supporting unlimited user customization.

Highway Crossing 3D Game | *OpenGL, C#*

January 2022 – May 2022

- Designed and implemented a 3D highway crossing simulation with audio effects and textured 3D models for immersive gameplay.
- Optimized collision detection and OpenGL rendering pipeline, reducing latency by 20% and improving responsiveness.

HONORS AND AWARDS

- Finalist - **Madhacks Fall 2023** - Selected among top teams for innovative software solutions in competitive hackathon environment.
- Finalist - **MadData Fall 2024** - Recognized for excellence in data analysis and machine learning application.